Assesment - 1

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Course : compiler Design
Course code : CSA1406
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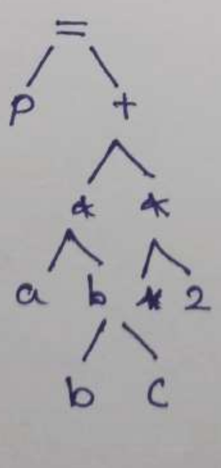
1) compiler Phases for Expression

$$P = (a*b) + (b*c)*2$$

a) Tokens generated during lexical analysis

Lexeme	Token
P	Identifer
=	Assignment operator
(	Left parenthesis
a	Identifer
*	Multiplication operator
b	Identifier
)	Right parenthesis
2	constant

b) syntax Tree



c) Intermediate code (Three address code)

$t_1 = a * b$

$t_2 = b * c$

$t_3 = t_2 * 2$

$t_4 = t_1 + t_3$

$P = t_4$

d) optimized Intermediate code

$t_1 = a * b$

$t_2 = b * c$

$t_2 = t_2 << 1$

$P = t_1 + t_2$

e) significance of final Target code

* Machine code generated for processor execution
* optimized for speed and memory.
* can be directly executed by hardware.

2) Five valid strings

a) $(aa^*)^*$

Exa:

1. a
2. a a
3. a a a
4. a a a a
5. a a a a a

b) $(b^*)abb$

Exa:

1. abb
2. babb
3. bbabb
4. bbbabb
5. bbbbabb

c) $(ab)aba(a+b^*)^*$

Exa:

1. aba
2. abaa
3. abab
4. aababa
5. bbabba

d) $(ab)^*$

Exa:

1. ε
2. ab
3. abab
4. ababab
5. abababab

e) $(aba)^* + (ab)^*$

Exa:

1. ε
2. aba
3. abaaba
4. aaa
5. bbb

i) Number of Tokens

```
main ()
{
  int a=10;
  char b = "abc";
  int c = 30;
  char d = 'xyz";
  m = 40.5;
}
```

Total Tokens = 25

ii) Lexical Errors

```
void main()
{
  int x = 10, y=20;
  char *a;
  a = &x;
  x = 1xab;
}
```

Errors:

1. a = &x; → Assiging address of integer variable to character pointer

2. 1xab
   * Invalid numeric literal
   * Identifer cannot start with a digit

4) Regular Expressions over {0,1}

i) Even number of 0's

$1^*(01^*01^*)^*$

ii) At least one 1

$(0+1)^*1(0+1)^*$

iii) Number of 0's and 1's are both odd

$(00+11)^*(01+10)(00+11)^*$

iv) start with 1 and End with 0

$1(0+1)^*0$

v) No consecutive 1's

$(0+10)^*(1+\epsilon)$

5) Lexical Analyzer.

Input:

```
int var123 = 12345;
float var456 = 123.45;
if (var123 == var456)
{
return;
}
```

a) Number of Tokens.

Total Tokens = 19

b) Categorized Tokens

Keywords
- int
- float
- if
- return

Identifiers
- var123
- var456
- var123
- var456

Operators
- =
- =
- ==

Separators
- ;
- ;
- (
-)
- {
- ;
- }

Integer literal

12345

Floating point literal

123.45

c) Estimated FSM state Transitions

Total estimated transitions:

int → 3

var123 → 6

= → 1

12345 → 5

; → 1

float → 5

var456 → 6

= → 1

123.45 → 6

; → 1

if → 2

(→ 1

var123 → 6

== → 2

var456 → 6

) → 1

{ → 1

return → 6

; → 1

} → 1

Total FSM transitions ≈ 61